

CHAPTER 1: INTRODUCTION

1.1 Introduction:

This project focuses on the development of a Multiple Disease Prediction System using machine learning in Python, and its deployment through the user-friendly Streamlit framework. By leveraging a comprehensive dataset containing patient information, medical history, and disease diagnoses, this system employs various machine learning models to predict the likelihood of multiple diseases. The system's performance is rigorously evaluated, ensuring accurate and reliable predictions.

The Streamlit deployment provides a user-friendly web interface that allows individuals to input their personal data and medical history, and in return, receive predictions for multiple diseases, along with associated probabilities. This system serves as a valuable tool for both healthcare professionals and individuals, empowering them with the ability to make informed decisions about their health. It demonstrates the potential of machine learning in healthcare and the practicality of accessible web interfaces in delivering predictive healthcare solutions.

1.2 Existing System and Limitations of Existing System:

Existing System:

- Manual Diagnostic Practices:
The current healthcare system relies heavily on manual diagnostic practices, leading to delays in disease detection.
- Limited Predictive Capabilities:
Predictive tools are underutilized, and there's a lack of comprehensive systems for predicting multiple diseases.
- Fragmented Health Records:
Health records are often fragmented across different systems, hindering a holistic view for accurate predictions.
- Insufficient Integration of Technology:
Limited integration of advanced technologies like machine learning and predictive analytics in routine healthcare practices.

- Patient Engagement Challenges:

Limited patient engagement in the diagnostic process, resulting in missed opportunities for early intervention.

Limitations of the Existing System:

- Manual Error and Subjectivity:

The manual diagnostic process is prone to human error and subjectivity, impacting the accuracy of disease predictions.

- Delayed Intervention:

Lack of predictive capabilities leads to delayed intervention, affecting treatment outcomes and patient well-being.

- Data Silos and Interoperability Issues:

Fragmented health records and interoperability challenges impede a comprehensive understanding of patient health.

- Underutilization of Advanced Technologies:

Limited integration and utilization of advanced technologies hamper the potential for early disease prediction.

- Inadequate Patient Involvement:

Insufficient patient involvement in the diagnostic process results in a missed opportunity for personalized healthcare management.

1.3 Need For The Proposed System:

- Early Disease Detection: Early detection of diseases is crucial for effective treatment and improved patient outcomes. Machine learning models can analyze extensive medical data to identify potential health risks and diseases at an early stage, facilitating timely intervention and treatment.
- Accessible Healthcare: With the growing demand for accessible and user-friendly healthcare solutions, deploying such a system via Streamlit offers an intuitive and

convenient platform for individuals to assess their health risks and make informed decisions.

- Data-Driven Healthcare: The increasing availability of healthcare data necessitates the utilization of advanced analytics and machine learning techniques. This system bridges the gap between data-driven healthcare and practical use by providing predictions and insights to both healthcare professionals and individuals.
- Personalized Medicine: Tailoring healthcare recommendations and interventions to an individual's specific risk factors and medical history is a key trend in modern medicine. This system enables personalized disease prediction, contributing to more effective and efficient healthcare delivery.

CHAPTER 2: PROPOSED SYSTEM

2.1 Problem Statement:

- **Predictive Healthcare Gaps:**

Challenge: Current healthcare lacks predictive tools, delaying disease detection and intervention.

- **Data Complexity and Integration:**

Challenge: Diverse patient data sources pose integration challenges for accurate disease prediction models.

- **User Accessibility in Healthcare Predictions:**

Challenge: Healthcare professionals and individuals face barriers in accessing and interpreting predictive models.

- **Integration of Multidimensional Data:**

Challenge: Incorporating and harmonizing multidimensional patient data for comprehensive disease predictions.

- **Real-World Applicability:**

Challenge: Bridging the gap between predictive models and practical healthcare decision-making for impactful results.

2.2 Objectives of Proposed System:

- **To Develop a Disease Prediction System:** The primary objective of this project is to create a disease prediction system that leverages machine learning to predict the likelihood of multiple diseases based on medical data and health records.
- **To Enable Early Detection:** One of the key aims is to enable early detection of diseases, which can lead to timely intervention and improved patient outcomes.
- **To Create a User-Friendly Interface:** I plan to design an intuitive web-based interface using Streamlit, which will allow healthcare professionals and individuals to easily input data and access prediction results.

- **To Enhance Healthcare Decision-Making:** The project aims to provide healthcare professionals with valuable insights to make informed decisions about patient care and treatment plans.
- **To Achieve Personal Learning Outcomes:** Ultimately, my goal is to achieve personal learning outcomes by gaining practical experience in data science, machine learning, web development, and project management.

2.3 Functional and Non-Functional Requirements:

Functional Requirements:

- Data Input and Processing:
The system should allow users to input personal and medical data, and it must process this data efficiently for predictive analysis.
- Disease Prediction Models:
Implement machine learning models for the accurate prediction of multiple diseases, including heart disease, diabetes, Parkinson's disease, and breast cancer.
- Predictive Analytics Dashboard:
Provide a user-friendly dashboard displaying disease predictions, associated probabilities, and relevant insights for both healthcare professionals and individuals.
- Privacy and Security Measures:
Implement robust measures to ensure the privacy and security of sensitive health information, complying with healthcare regulations.
- User Feedback and Interaction:
Include features for users to provide feedback on predictions and interact with the system for improved user engagement.

Non-Functional Requirements:

- Performance:
The system must handle a substantial amount of data and user requests, ensuring fast response times for efficient user interactions.
- Scalability:
Design the system to be scalable, accommodating an increasing number of users and growing datasets.

- Usability:
The user interface must be intuitive, ensuring ease of use for both healthcare professionals and individuals without requiring extensive training.
- Interoperability:
Ensure compatibility and interoperability with existing healthcare systems and data standards for seamless integration into healthcare practices.
- Maintainability:
Design the system with modular components for ease of maintenance and updates to incorporate advancements in predictive analytics and healthcare technologies.

2.4 Scope of System:

- Disease Prediction: The primary scope of the system is to predict the likelihood of multiple diseases based on user-provided information and medical data. This includes the ability to predict a wide range of diseases, from common chronic conditions to less prevalent health issues.
- User Accessibility: The system aims to provide a user-friendly interface, making it accessible to both healthcare professionals and individuals seeking to assess their health risks. This accessibility is essential to reach a diverse user base.
- Data Integration: The system have the capability to integrate and analyze diverse healthcare data sources, including patient demographics, medical history, symptoms, and diagnostic records, to make accurate predictions.
- Machine Learning Models: It encompasses the development and selection of machine learning models, such as support vector machines, logistic regression to ensure accurate and reliable disease predictions.

2.5 Module Specification:

1. Heart Disease Prediction Module

Objective: Develop a module for predicting heart disease risk.

Submodules:

Data Collection: Gather relevant medical data for heart disease prediction, including patient profiles and diagnostic records.

Data Preprocessing: Clean and preprocess the data to handle missing values, outliers, and ensure data quality.

Model Development: Create a machine learning model tailored to predict heart disease, considering key features and diagnostic indicators.

Model Training and Evaluation: Train the heart disease prediction model using historical data and evaluate its performance with appropriate metrics.

Integration with Streamlit: Integrate the heart disease prediction model into the Streamlit interface, allowing user interaction and results display.

2. Diabetes Prediction Module

Objective: Develop a module for predicting the likelihood of diabetes.

Submodules:

Data Collection: Collect comprehensive medical data related to diabetes risk, including patient profiles, blood tests, and lifestyle factors.

Data Preprocessing: Clean, preprocess, and transform the data for effective machine learning model input.

Model Development: Build a machine learning model specialized in predicting diabetes, considering relevant patient attributes.

Model Training and Evaluation: Train and rigorously evaluate the diabetes prediction model, using performance metrics such as accuracy, precision, and recall.

Integration with Streamlit: Integrate the diabetes prediction model into the Streamlit interface to allow user input and results presentation.

3. Parkinson's Disease Prediction Module

Objective: Develop a module for predicting Parkinson's disease.

Submodules:

Data Collection: Gather patient data, diagnostic test results, and clinical assessments associated with Parkinson's disease.

Data Preprocessing: Prepare the data for machine learning by addressing missing values and normalizing data where necessary.

Model Development: Design a machine learning model optimized for predicting the likelihood of Parkinson's disease.

Model Training and Evaluation: Train and evaluate the Parkinson's disease prediction model with a focus on achieving high accuracy.

Integration with Streamlit: Integrate the Parkinson's disease prediction model into the Streamlit interface for user interaction.

4. Breast Cancer Prediction Module

Objective: Create a module for predicting breast cancer risk.

Submodules:

Data Collection: Acquire diverse datasets, including mammography results, genetic profiles, and patient medical history, to predict breast cancer.

Data Preprocessing: Clean, preprocess, and format the data for effective machine learning input.

Model Development: Construct a dedicated machine learning model for breast cancer prediction, emphasizing feature selection and importance.

Model Training and Evaluation: Train and rigorously evaluate the breast cancer prediction model, considering metrics such as accuracy and sensitivity.

Integration with Streamlit: Integrate the breast cancer prediction model into the Streamlit interface for data input and result presentation.

5. Streamlit Web Interface Module

Objective: Develop a user-friendly web interface using Streamlit for data input and result presentation.

Submodules:

Data Input Forms: Create user-friendly and intuitive forms for users to input their health data securely.

Prediction Results Display: Design an easy-to-understand display for presenting prediction results for each disease, providing users with valuable insights.

2.6 Operating Environment- Hardware And Software

Hardware Requirements:

- **Personal Computer:** You can use your personal computer or a laptop for development and deployment, as long as it meets the software requirements.
- **Standard Hardware:** A standard laptop or desktop with a reasonable amount of RAM (4GB or more) and a modern CPU should suffice for small-scale experimentation.
- **Internet Connection:** A stable internet connection for accessing libraries and resources online.

Software Requirements:

- **Operating System:** Windows.

- Programming Language: Python is the primary language for this mini project.
- Python Libraries: Depending on the scope of my project, I am using Python libraries like scikit-learn, pandas, numpy, and Streamlit.
- Development Environment: An integrated development environment (IDE) like Spyder can help with coding and testing.
- Streamlit: Streamlit, a Python library for creating web applications, via pip.
- Machine Learning Libraries: Lightweight machine learning libraries like scikit-learn.

CHAPTER 3: REQUIREMENT DETERMINATION AND ANALYSIS

3.1 Fact Finding Methods:

Stakeholder Interviews:

Conduct interviews with healthcare professionals, data scientists, and potential users to understand their expectations, needs, and concerns regarding disease prediction.

Surveys on Healthcare Practices:

Distribute surveys to healthcare practitioners to gather insights into current diagnostic practices, challenges faced, and the perceived benefits of a predictive disease system.

Patient Data Analysis:

Analyze existing patient datasets to understand the types of data available, data quality, and potential challenges in integrating diverse health records.

Prototyping with Healthcare Professionals:

Develop a prototype of the disease prediction system and gather feedback from healthcare professionals to refine the system's features and functionalities.

Observation of Clinical Practices:

Observe healthcare professionals in clinical settings to understand their workflows, decision-making processes, and potential points of integration for predictive analytics.

Analysis of Existing Disease Prediction Models:

Review and analyze existing disease prediction models to understand the strengths, limitations, and potential enhancements for the proposed system.

3.2 Feasibility Study:

Introduction:

In this section, we will conduct a feasibility study to assess the viability of our mini project, which aims to develop a "Multiple Disease Prediction System." This study will help us understand whether our project is practical and achievable within the defined scope.

Technical Feasibility:

Data Availability:

I have confirmed the availability of relevant medical datasets for training and testing our machine learning models. The data is publicly accessible and suitable for our project's scope.

Technical Expertise:

As the developer of this mini project, I am learning the necessary technical skills, including proficiency in Python and basic machine learning concepts. I am confident in my ability to learn and apply additional skills required for this project.

Hardware and Software:

The hardware and software resources needed for this mini project are readily available on my personal computer. I can use my existing machine for development and deployment.

Operational Feasibility:

Resource Availability:

The resources required for this project, such as hardware, software, and data, are within reach and will be accessible throughout the project's duration.

Scalability:

I understand that this mini project's focus is on demonstrating feasibility within a limited scope, and scalability concerns are secondary.

Scheduling Feasibility:

Project Timeline:

I have created a project timeline that aligns with my academic schedule. This timeline allows me to meet milestones while balancing coursework and other commitments.

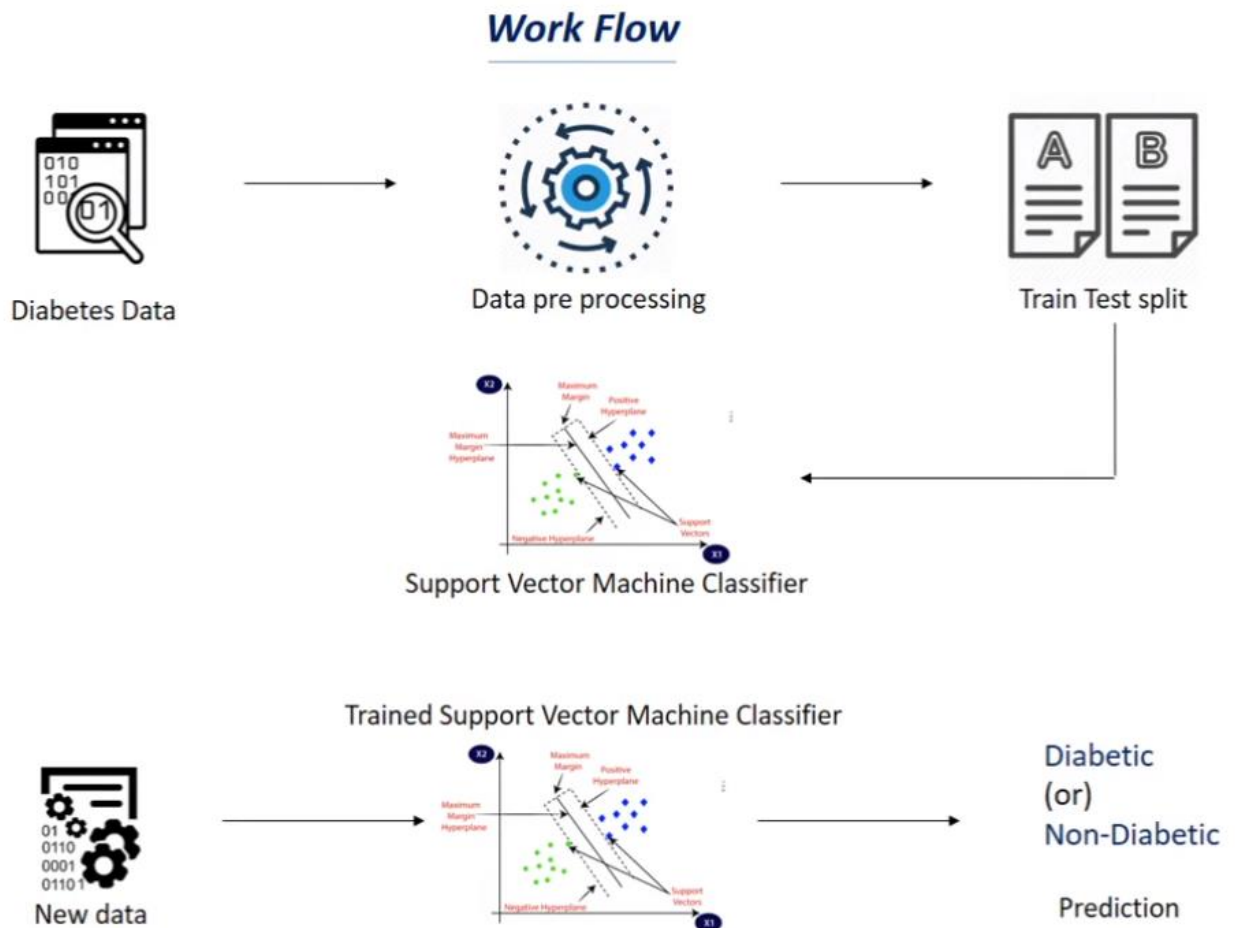
Conclusion:

In conclusion, the feasibility study indicates that our mini project is both technically and operationally feasible within the defined scope. I am confident in my ability to develop the Multiple Disease Prediction System and complete it successfully.

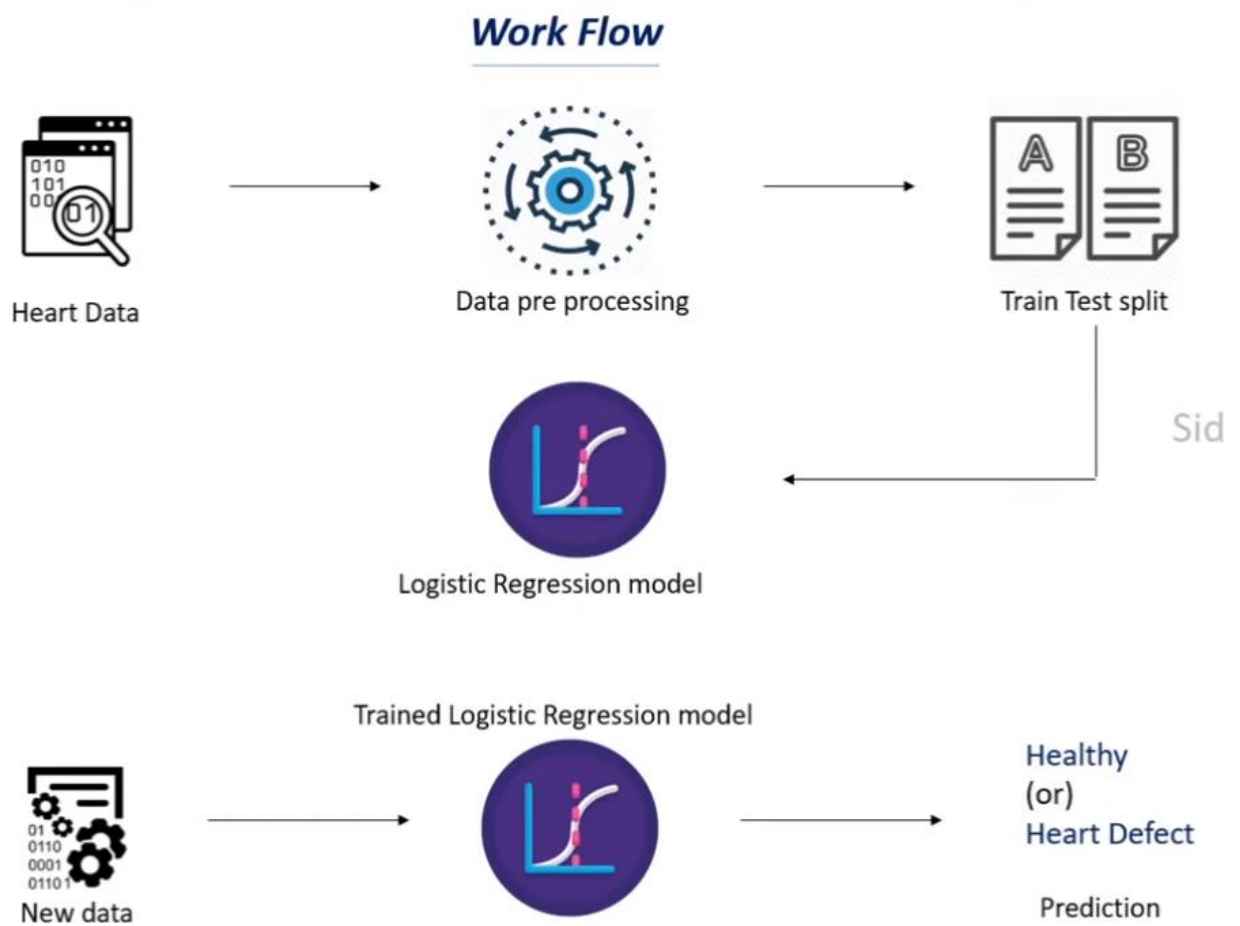
CHAPTER 4: SYSTEM ANALYSIS AND DESIGN

4.1 Workflow Diagram:

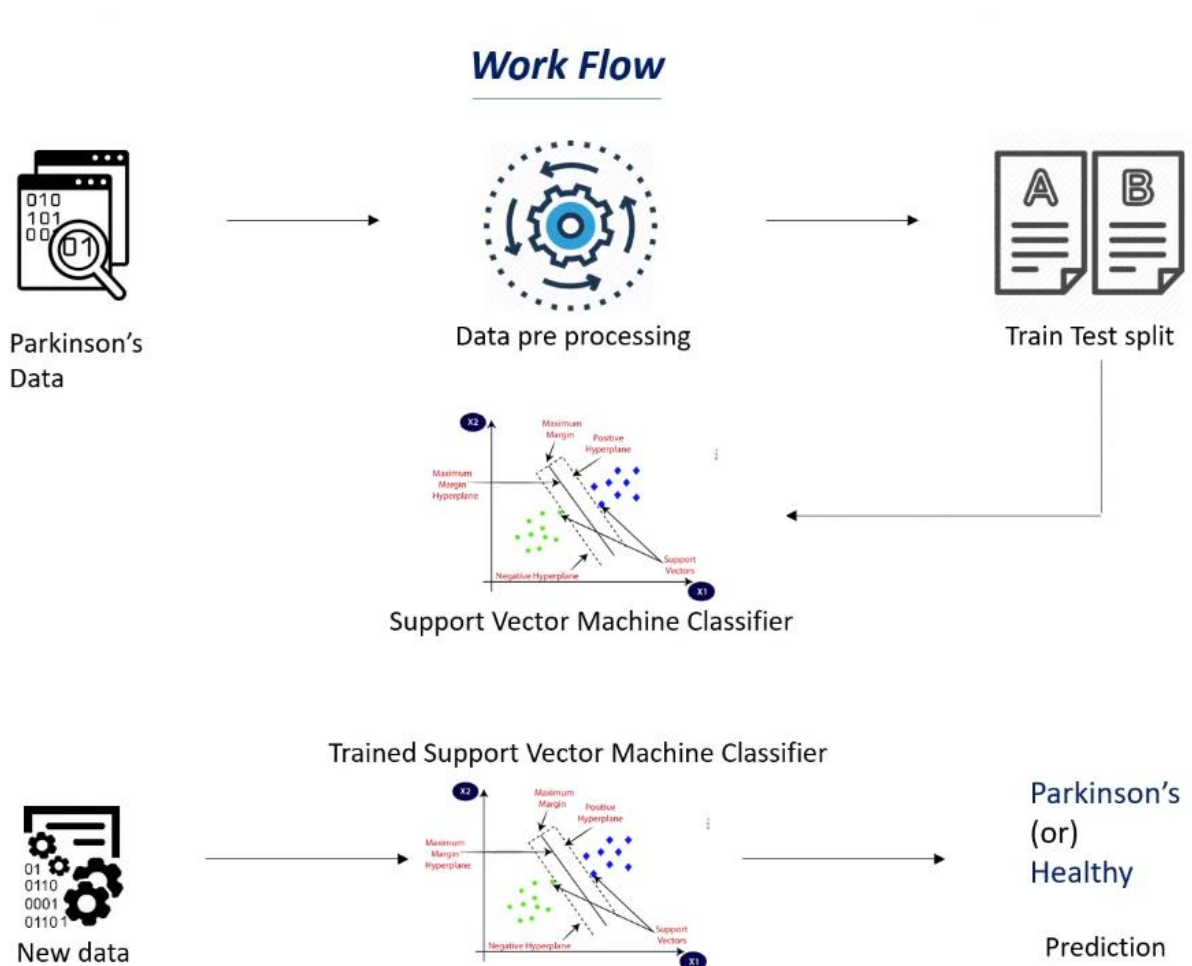
1. Diabetes Prediction Model Work Flow:



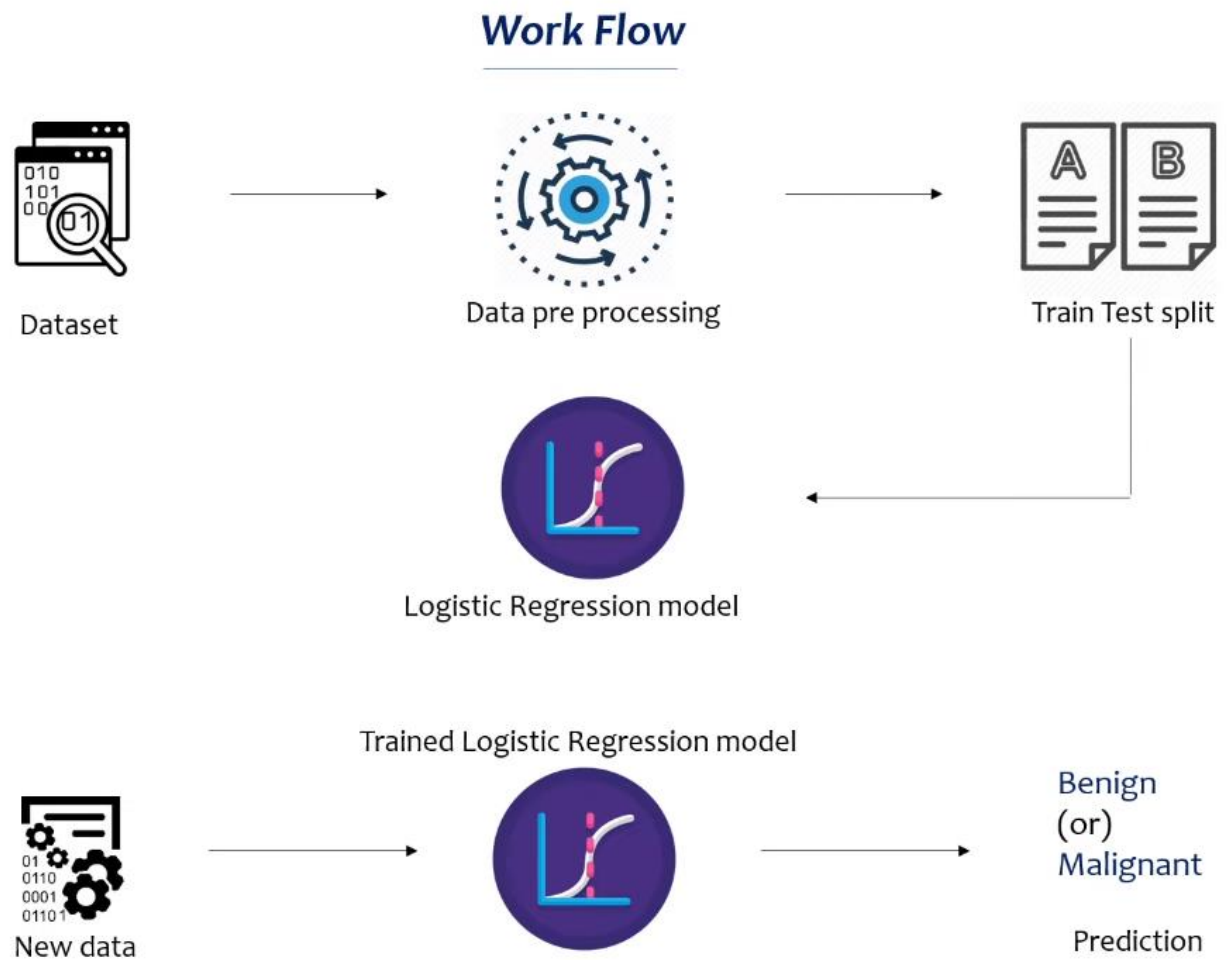
2. Heart Disease Prediction Model Work Flow:



1. Parkinson's Prediction Model Work Flow:

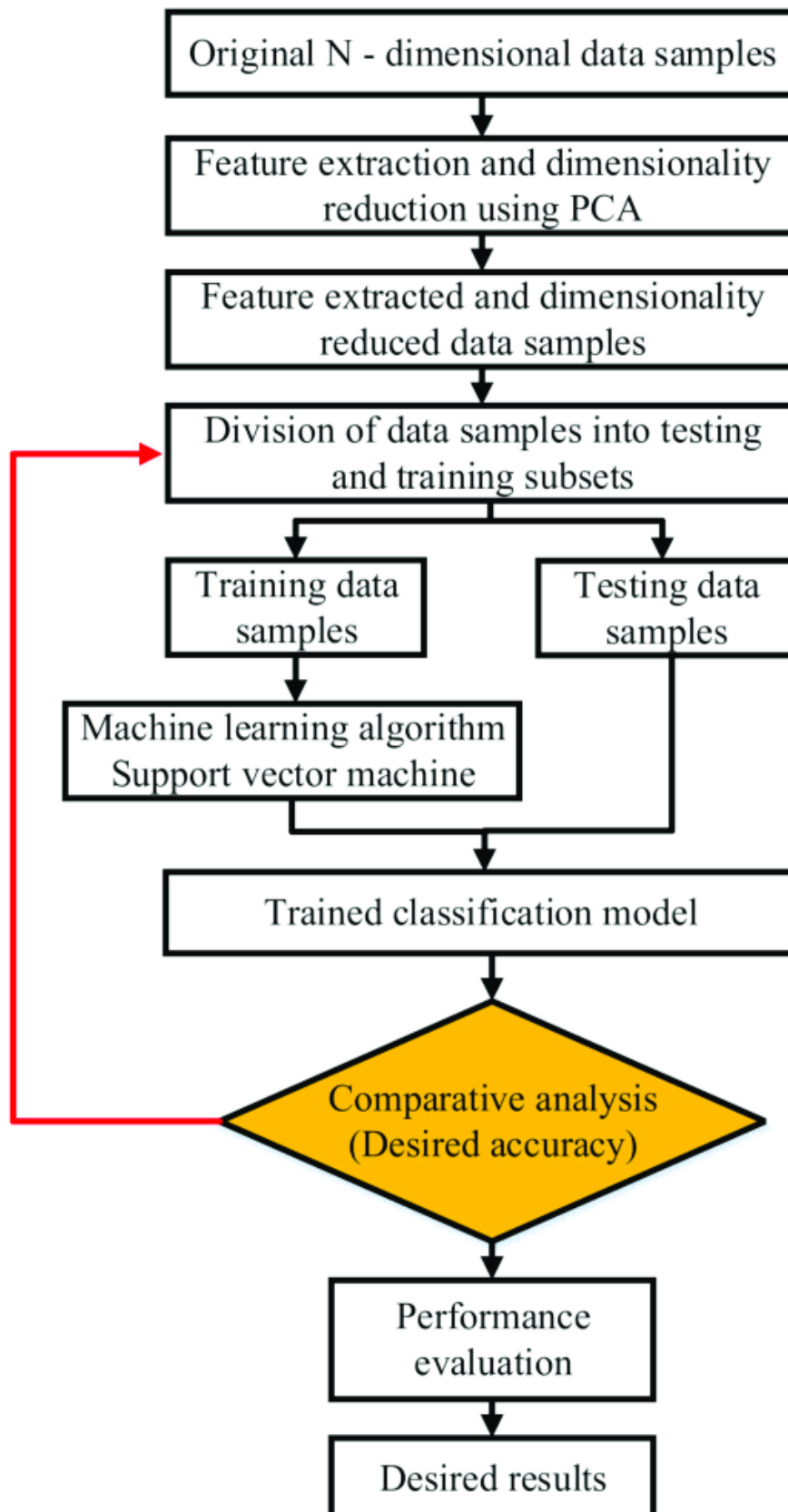


1. Breast Cancer Prediction Model Work Flow:

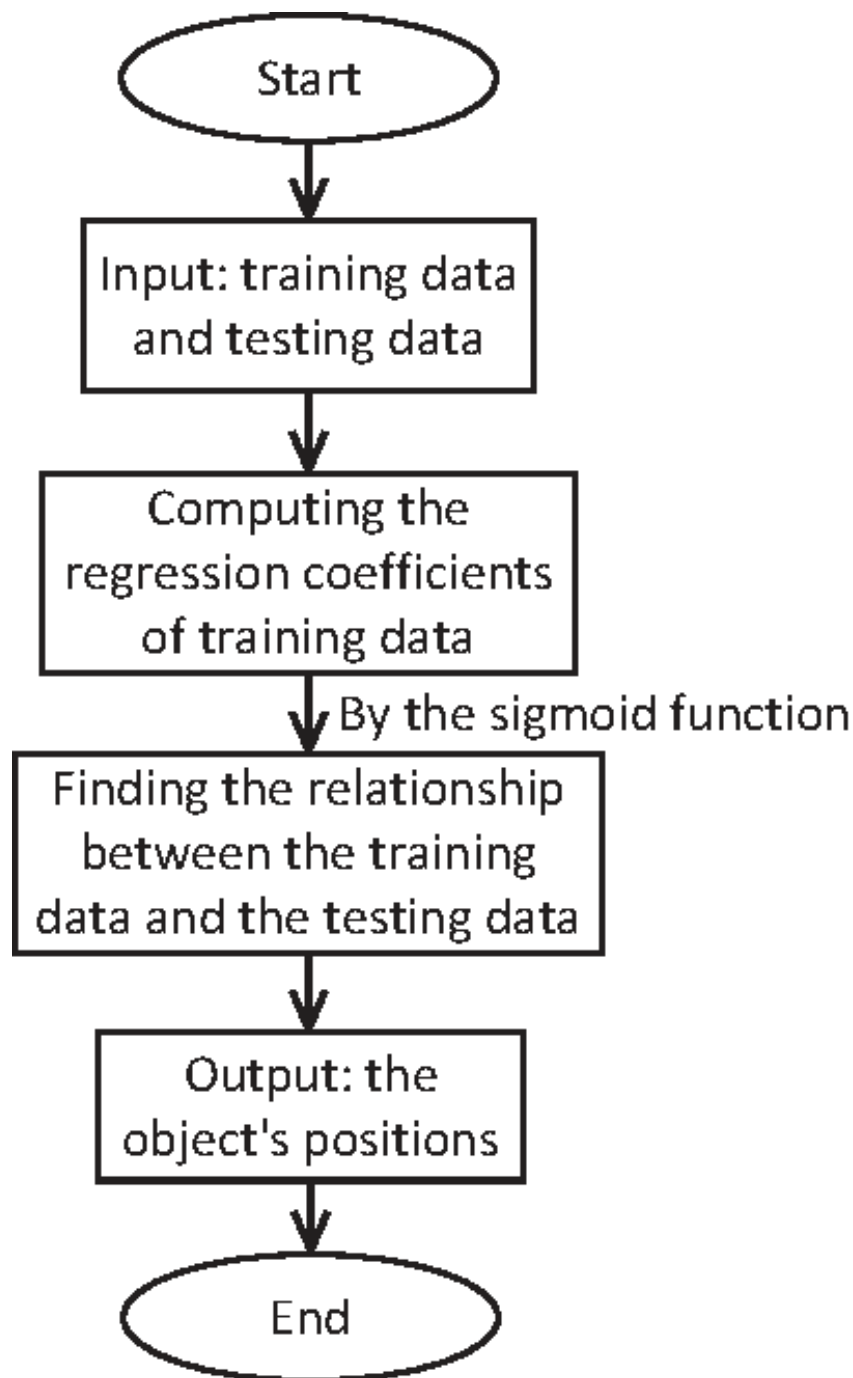


4.2 Flowchart Diagram:

1. Support Vector Machine algorithm:



2. Logistic Regression Algorithm:



4.3 USER INTERFACE DESIGN:

1. Diabetes Prediction Model:

The screenshot shows a web browser at localhost:8501 displaying the 'Diabetes Prediction' interface. On the left is a sidebar titled 'Multiple Disease Prediction System' with four options: 'Diabetes Prediction' (highlighted in red), 'Heart Disease Prediction', 'Parkinsons Prediction', and 'Breast Cancer Prediction'. The main area is titled 'Diabetes Prediction' and contains a form with the following fields, all with the value '1': 'Number of Pregnancies', 'Glucose Level', 'Blood Pressure value', 'Skin Thickness value', 'Insulin Level', 'BMI value', 'Diabetes Pedigree Function value', and 'Age of the Person'. Below the form is a 'Diabetes Test Result' button. The result area shows a green box with the text 'The person is not diabetic'.

2. Heart Disease Prediction Model:

The screenshot shows the same web browser at localhost:8501, but now displaying the 'Heart Disease Prediction' interface. The sidebar is identical, but 'Heart Disease Prediction' is now highlighted in red. The main area is titled 'Heart Disease Prediction' and contains a form with the following fields, all with the value '1': 'Age', 'Sex', 'Chest Pain types', 'Resting Blood Pressure', 'Serum Cholesterol in mg/dl', 'Fasting Blood Sugar > 120 mg/dl', 'Resting Electrocardiographic results', 'Maximum Heart Rate achieved', 'Exercise Induced Angina', 'ST depression induced by exercise', 'Slope of the peak exercise ST segment', 'Major vessels colored by flourosopy', and a text input field for 'thal: 0 = normal; 1 = fixed defect; 2 = reversable defect'. Below the form is a 'Heart Disease Test Result' button. The result area shows a green box with the text 'The person does not have any heart disease'.

3. Parkinson's Disease Prediction Model:

The screenshot shows a web browser window with the URL `localhost:8501`. The application is titled "Parkinson's Disease Prediction". On the left, a sidebar menu for the "Multiple Disease Prediction System" includes options for Diabetes, Heart Disease, Parkinson's (selected), and Breast Cancer. The main area contains a grid of 15 input fields, each with a value of "1". The inputs are: MDVP: Fo(Hz), MDVP: Fhi(Hz), MDVP: Flo(Hz), MDVP: Jitter(%), MDVP: Jitter(Abs), MDVP: RAP, MDVP: PPQ, Jitter: DDP, MDVP: Shimmer, MDVP: Shimmer(dB), Shimmer: APQ3, Shimmer: APQ5, MDVP: APQ, Shimmer: DDA, NHR, HNR, RPDE, DFA, spread1, spread2, D2, and PPE. Below the inputs is a "Parkinson's Test Result" button. A green message box at the bottom states "The person has Parkinson's disease".

MDVP: Fo(Hz)	MDVP: Fhi(Hz)	MDVP: Flo(Hz)	MDVP: Jitter(%)	MDVP: Jitter(Abs)
1	1	1	1	1
MDVP: RAP	MDVP: PPQ	Jitter: DDP	MDVP: Shimmer	MDVP: Shimmer(dB)
1	1	1	1	1
Shimmer: APQ3	Shimmer: APQ5	MDVP: APQ	Shimmer: DDA	NHR
1	1	1	1	1
HNR	RPDE	DFA	spread1	spread2
1	1	1	1	1
D2	PPE			
1	1			

Parkinson's Test Result

The person has Parkinson's disease

4. Breast Cancer Prediction Model:

The screenshot shows the same web browser window, but the application is now titled "Breast Cancer Prediction". The sidebar menu is identical, but "Breast Cancer Prediction" is selected. The main area contains a grid of 20 input fields, each with a value of "1". The inputs are: Radius: Mean, Texture: Mean, Perimeter: Mean, Area: Mean, Smoothness: Mean, Radius: Standard Error, Texture: Standard Error, Perimeter: Standard Error, Area: Standard Error, Smoothness: Standard Error, Radius: Worst, Texture: Worst, Perimeter: Worst, Area: Worst, Smoothness: Worst, Compactness: Mean, Concavity: Mean, Concave Points: Mean, Symmetry: Mean, Fractal Dimension: Mean, Compactness: Standard Error, Concavity: Standard Error, Concave Points: Standard Error, Symmetry: Standard Error, Fractal Dimension: Standard Error, Compactness: Worst, Concavity: Worst, Concave Points: Worst, Symmetry: Worst, and Fractal Dimension: Worst. Below the inputs is an "Evaluate Test Result" button.

Radius: Mean	Texture: Mean	Perimeter: Mean	Area: Mean	Smoothness: Mean
1	1	1	1	1
Radius: Standard Error	Texture: Standard Error	Perimeter: Standard Error	Area: Standard Error	Smoothness: Standard Error
1	1	1	1	1
Radius: Worst	Texture: Worst	Perimeter: Worst	Area: Worst	Smoothness: Worst
1	1	1	1	1
Compactness: Mean	Concavity: Mean	Concave Points: Mean	Symmetry: Mean	Fractal Dimension: Mean
1	1	1	1	1
Compactness: Standard Error	Concavity: Standard Error	Concave Points: Standard Error	Symmetry: Standard Error	Fractal Dimension: Standard Error
1	1	1	1	1
Compactness: Worst	Concavity: Worst	Concave Points: Worst	Symmetry: Worst	Fractal Dimension: Worst
1	1	1	1	1

Evaluate Test Result

CHAPTER 5: DRAWBACKS AND LIMITAION

Data Quality and Availability:

Drawback: The accuracy and reliability of predictions heavily rely on the quality and availability of diverse patient data.

Limitation: Incomplete or inaccurate data can lead to biased predictions and reduced system effectiveness.

Ethical and Privacy Concerns:

Drawback: Predictive systems dealing with sensitive health information raise ethical concerns related to patient privacy and consent.

Limitation: Striking a balance between effective prediction and ensuring patient privacy is challenging.

Algorithm Bias and Interpretability:

Drawback: Machine learning algorithms may exhibit bias, potentially leading to disparities in disease predictions across demographic groups.

Limitation: Ensuring algorithm transparency and interpretability is challenging, impacting user trust and acceptance.

Dynamic Nature of Health Data:

Drawback: Health data is dynamic and subject to constant changes, making it challenging to maintain accurate predictive models over time.

Limitation: Regular model updates and adaptation to evolving health trends are necessary but resource-intensive.

User Engagement and Adherence:

Drawback: The success of the system relies on user engagement, and there may be challenges in ensuring consistent user adherence to inputting accurate and timely data.

Limitation: Incomplete or sporadic data input can affect the reliability of predictions.

CHAPTER 6: PROPOSED ENHANSMENT

Data Quality Assurance:

Enhancement: Implement advanced data quality assurance measures, including data cleansing algorithms and outlier detection, to enhance the accuracy and reliability of predictive models.

Real-time Data Integration:

Enhancement: Implement mechanisms for real-time integration of health data to ensure the system adapts to dynamic changes promptly, improving the currency and relevance of predictions.

User-Centric Design and Engagement:

Enhancement: Employ user-centric design principles to enhance the system's usability. Implement features such as personalized health insights, reminders, and incentives to encourage consistent user engagement.

Continuous Model Training:

Enhancement: Implement a continuous model training approach using updated datasets, ensuring that the predictive models evolve with changing health trends and maintain high accuracy over time.

Ethical Framework and Explainability:

Enhancement: Develop a transparent ethical framework that outlines principles for patient consent, data use, and algorithmic decision-making. Enhance model interpretability to address bias concerns.

CHAPTER 7: CONCLUSTION

In conclusion, the development of the Multiple Disease Prediction System represents a pivotal stride towards transforming healthcare through innovative predictive analytics. The fact-finding methods, encompassing stakeholder interviews, surveys, and data analysis, have provided invaluable insights into the complexities of healthcare processes, data dynamics, and user expectations. The proposed enhancements, rooted in addressing drawbacks and leveraging advancements, strive to forge a resilient system that not only accurately predicts diseases but also places a strong emphasis on user engagement, transparency, and privacy controls.

By acknowledging and actively mitigating challenges such as data quality concerns and algorithmic biases, the envisioned system seeks to set new standards in predictive healthcare. The integration of disease-specific models, continuous training mechanisms, and real-time data updates demonstrates a commitment to staying abreast of evolving health trends. Moving forward, collaboration with healthcare professionals, users, and regulatory bodies will be paramount to ensuring that the system aligns with emerging healthcare standards and remains ethically robust.

In essence, the Multiple Disease Prediction System is poised to be a transformative force in healthcare. By amalgamating sophisticated analytics with user-centric design, the system endeavors to empower individuals, facilitate early intervention, and showcase the profound potential of technology in shaping the future landscape of healthcare. This endeavor marks a significant leap towards a more proactive, personalized, and effective healthcare paradigm.

CHAPTER 8: BIBLOGRAPHY

MY HELPING HANDS:

- WIKIPEDIA
- JAVATPOINT
- GEEGKFORGEEKS
- YOUTUBE
- STACKOVERFLOW
- KAGGLE
- MY PROJECT GUIDE

CHAPTER 9: SAMPLE CODE

Code for Developing and Deploying Web Application using Streamlit:

```
import pickle

import streamlit as st

from streamlit_option_menu import option_menu


# loading the saved models

diabetes_model = pickle.load(open('D:/Multiple_Disease_Prediction/saved
models/diabetes_model.sav', 'rb'))


heart_disease_model = pickle.load(open('D:/Multiple_Disease_Prediction/saved
models/heart_disease_model.sav','rb'))


parkinsons_model = pickle.load(open('D:/Multiple_Disease_Prediction/saved
models/parkinsons_model.sav', 'rb'))


breast_cancer_model = pickle.load(open('D:/Multiple_Disease_Prediction/saved
models/breast_cancer_model.sav', 'rb'))


# sidebar for navigation

with st.sidebar:

    selected = option_menu('Multiple Disease Prediction System',
                           ['Diabetes Prediction',
                            'Heart Disease Prediction',
                            'Parkinsons Prediction',
                            'Breast Cancer Prediction'],
                           icons=['activity','heart','person','gender-female'],
                           default_index=0)
```

```

# Diabetes Prediction Page

if (selected == 'Diabetes Prediction'):

    # page title
    st.title('Diabetes Prediction ')

    # getting the input data from the user
    col1, col2, col3 = st.columns(3)

    with col1:
        Pregnancies = st.text_input('Number of Pregnancies')

    with col2:
        Glucose = st.text_input('Glucose Level')

    with col3:
        BloodPressure = st.text_input('Blood Pressure value')

    with col1:
        SkinThickness = st.text_input('Skin Thickness value')

    with col2:
        Insulin = st.text_input('Insulin Level')

    with col3:
        BMI = st.text_input('BMI value')

    with col1:
        DiabetesPedigreeFunction = st.text_input('Diabetes Pedigree Function value')

```

```

with col2:

    Age = st.text_input('Age of the Person')


# code for Prediction
diab_diagnosis = "

# creating a button for Prediction
if st.button('Diabetes Test Result'):

    diab_prediction = diabetes_model.predict([[Pregnancies, Glucose, BloodPressure,
SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age]])

    if (diab_prediction[0] == 1):
        diab_diagnosis = 'The person is diabetic'
    else:
        diab_diagnosis = 'The person is not diabetic'

st.success(diab_diagnosis)


# Heart Disease Prediction Page
if (selected == 'Heart Disease Prediction'):

    # page title
    st.title('Heart Disease Prediction ')

    col1, col2, col3 = st.columns(3)

    with col1:

        age = st.text_input('Age')

    with col2:

```

```
sex = st.text_input('Sex')

with col3:

    cp = st.text_input('Chest Pain types')

with col1:

    trestbps = st.text_input('Resting Blood Pressure')

with col2:

    chol = st.text_input('Serum Cholestoral in mg/dl')

with col3:

    fbs = st.text_input('Fasting Blood Sugar > 120 mg/dl')

with col1:

    restecg = st.text_input('Resting Electrocardiographic results')

with col2:

    thalach = st.text_input('Maximum Heart Rate achieved')

with col3:

    exang = st.text_input('Exercise Induced Angina')

with col1:

    oldpeak = st.text_input('ST depression induced by exercise')

with col2:

    slope = st.text_input('Slope of the peak exercise ST segment')

with col3:
```

```

ca = st.text_input('Major vessels colored by flourosopy')

with col1:

    thal = st.text_input('thal: 0 = normal; 1 = fixed defect; 2 = reversable defect')


# code for Prediction
heart_diagnosis = "

# creating a button for Prediction

if st.button('Heart Disease Test Result'):

    heart_prediction = heart_disease_model.predict([[age, sex, cp, trestbps, chol, fbs,
restecg,thalach,exang,oldpeak,slope,ca,thal]])

    if (heart_prediction[0] == 1):
        heart_diagnosis = 'The person is having heart disease'
    else:
        heart_diagnosis = 'The person does not have any heart disease'

st.success(heart_diagnosis)


# Parkinson's Prediction Page
if (selected == "Parkinsons Prediction"):
    # page title
    st.title("Parkinson's Disease Prediction ")
    col1, col2, col3, col4, col5 = st.columns(5)

    with col1:
        fo = st.text_input('MDVP: Fo(Hz)')

```

with col2:

fhi = st.text_input('MDVP: Fhi(Hz)')

with col3:

flo = st.text_input('MDVP: Flo(Hz)')

with col4:

Jitter_percent = st.text_input('MDVP: Jitter(%)')

with col5:

Jitter_Abs = st.text_input('MDVP: Jitter(Abs)')

with col1:

RAP = st.text_input('MDVP: RAP')

with col2:

PPQ = st.text_input('MDVP: PPQ')

with col3:

DDP = st.text_input('Jitter: DDP')

with col4:

Shimmer = st.text_input('MDVP: Shimmer')

with col5:

Shimmer_dB = st.text_input('MDVP: Shimmer(dB)')

with col1:

APQ3 = st.text_input('Shimmer: APQ3')

with col2:

```
APQ5 = st.text_input('Shimmer: APQ5')
```

with col3:

```
APQ = st.text_input('MDVP: APQ')
```

with col4:

```
DDA = st.text_input('Shimmer: DDA')
```

with col5:

```
NHR = st.text_input('NHR')
```

with col1:

```
HNR = st.text_input('HNR')
```

with col2:

```
RPDE = st.text_input('RPDE')
```

with col3:

```
DFA = st.text_input('DFA')
```

with col4:

```
spread1 = st.text_input('spread1')
```

with col5:

```
spread2 = st.text_input('spread2')
```

with col1:

```
D2 = st.text_input('D2')
```

```

with col2:

    PPE = st.text_input('PPE')

# code for Prediction
parkinsons_diagnosis = "

# creating a button for Prediction
if st.button("Parkinson's Test Result"):

    parkinsons_prediction = parkinsons_model.predict([[fo, fhi, flo, Jitter_percent,
Jitter_Abs, RAP,
PPQ,DDP,Shimmer,Shimmer_dB,APQ3,APQ5,APQ,DDA,NHR,HNR,RPDE,DFA,spread1,s
pread2,D2,PPE]])

    if (parkinsons_prediction[0] == 1):
        parkinsons_diagnosis = "The person has Parkinson's disease"
    else:
        parkinsons_diagnosis = "The person does not have Parkinson's disease"

st.success(parkinsons_diagnosis)


#Breast Cancer Prediction
if (selected == "Breast Cancer Prediction"):
    # page title
    st.title("Breast Cancer Prediction")

    col1, col2, col3, col4, col5= st.columns(5)

    with col1:
        rad_m = st.text_input('Radius: Mean')

```



```
with col1:
    rad_se = st.text_input('Radius: Standard Error')

with col1:
    rad_w = st.text_input('Radius: Worst')

with col2:
    text_m = st.text_input('Texture: Mean')

with col2:
    text_se = st.text_input('Texture: Standard Error')

with col2:
    text_w = st.text_input('Texture: Worst')

with col3:
    peri_m = st.text_input('Perimeter: Mean')

with col3:
    peri_se = st.text_input('Perimeter: Standard Error')

with col3:
    peri_w = st.text_input('Perimeter: Worst')

with col4:
    area_m = st.text_input('Area: Mean')

with col4:
    area_se = st.text_input('Area:: Standard Errors..')
```

with col4:

area_w = st.text_input('Area: Worst')

with col5:

smth_m = st.text_input('Smoothness: Mean')

with col5:

smth_se = st.text_input('Smoothness: Standard Error')

with col5:

smth_w = st.text_input('Smoothness: Worst')

with col1:

comp_m = st.text_input('Compactness: Mean')

with col1:

comp_se = st.text_input('Compactness: Standard Error')

with col1:

comp_w = st.text_input('Compactness: Worst')

with col2:

con_m = st.text_input('Concavity: Mean')

with col2:

con_se = st.text_input('Concavity: Standard Error')

with col2:

con_w = st.text_input('Concavity: Worst')

```
with col3:
    cp_m = st.text_input('Concave Points: Mean')

with col3:
    cp_se = st.text_input('Concave Points: Standard Error')

with col3:
    cp_w = st.text_input('Concave Points: Worst')

with col4:
    sym_m = st.text_input('Symmetry: Mean')

with col4:
    sym_se = st.text_input('Symmetry: Standard Error')

with col4:
    sym_w = st.text_input('Symmetry: Worst')

with col5:
    fd_m = st.text_input('Fractal Dimension: Mean')

with col5:
    fd_se = st.text_input('Fractal Dimension: Standard Error')

with col5:
    fd_w = st.text_input('Fractal Dimension: Worst')

# code for Prediction
breast_cancer_diagnosis = "
```

```

# creating a button for Prediction

if st.button('Evaluate Test Result'):

    breast_cancer_prediction = breast_cancer_model.predict([[rad_m, text_m, peri_m,
area_m, smth_m, comp_m, con_m, cp_m, sym_m, fd_m, rad_se, text_se, peri_se, area_se,
smth_se, comp_se, con_se, cp_se, sym_se, fd_se, rad_w, text_w, peri_w, area_w, smth_w,
comp_w, con_w, cp_w, sym_w, fd_w]])

    if (breast_cancer_prediction[0] == 1):

        breast_cancer_diagnosis = 'Cancer Tissues are detected'

    else:

        breast_cancer_diagnosis = 'Cancer Tissues are not detected'

st.success(breast_cancer_diagnosis)

```